

# Electrical Machine I

**Course Code: EEE 2151**

## DC Generator

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# DC Generators

## Introduction

- DC generators are DC machines used as generators.
- There's no real difference between a DC generator or motor except for the direction of power flow.
- The various types of DC generators available differ in their terminal (voltage-current) characteristics and therefore in the applications to which they are suited.

- In this course, the following types DC generators will be discussed in details:
  - Separately excited DC generator
    - the field flux is derived from a separate power source and independent of the generator itself.
  - Shunt generator
    - the field flux is derived by connecting the field circuit directly across the terminals of the generator.

- DC generators are compared by their voltages, power ratings, efficiencies and voltage regulations defined as

$$V R = \frac{V_{nl} - V_{fl}}{V_{fl}} \times 100\%$$

$V_{nl}$  = no-load terminal voltage

$V_{fl}$  = full-load terminal voltage

- VR is the percentage change in terminal voltage when the load is removed from the generator.
- DC generators are driven by a source of mechanical power called the prime mover of the generator. It can be a steam engine, diesel engine or motor.

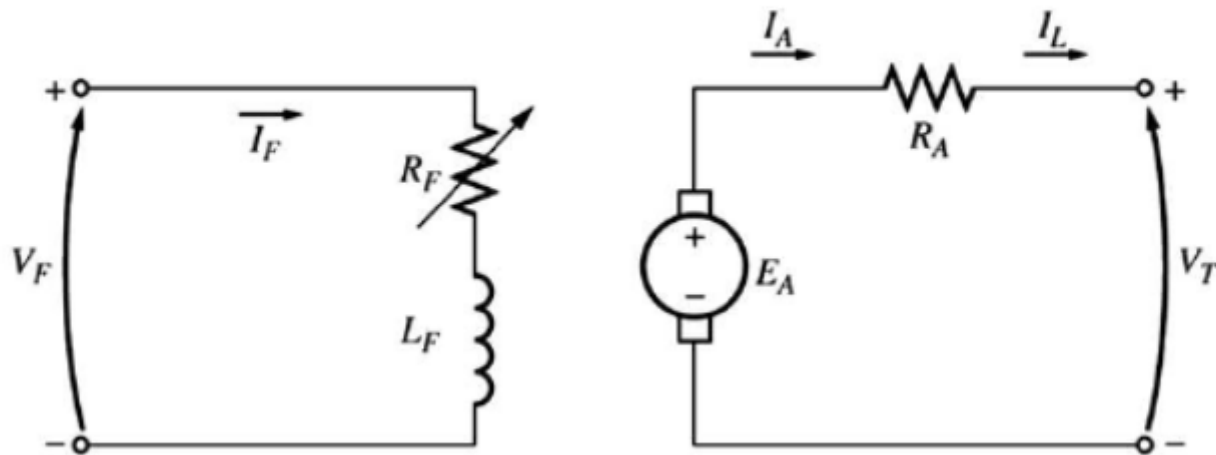
Example. A 100-kW, 1800 r/min DC generator operating at rated load has a terminal voltage of 240V. If the voltage regulation is 2.3 percent, determine the no-load voltage.

$$V R = \frac{V_{nl} - V_{fl}}{V_{fl}} \times 100\%$$

$$\Rightarrow 0.023 = \frac{V_{nl} - 240}{240}$$

$$\Rightarrow V_{nl} = 245.52 \text{ V}$$

## Separately Excited DC generator



Equivalent circuit of a separately excited generator

$R_F$  - combination of the resistance of the field windings and the external variable resistor.

$I_L$  - the current flowing in the lines connected to the terminals of the generator.

$$I_A = I_L, \quad V_T = E_A - I_A R_A, \quad I_F = \frac{V_F}{R_F}$$

Example. Determine the internal generated voltage, at rated conditions, of a separately excited generator which is rated at 25-kW, 250-V and 1450-r/min and has the following parameters.

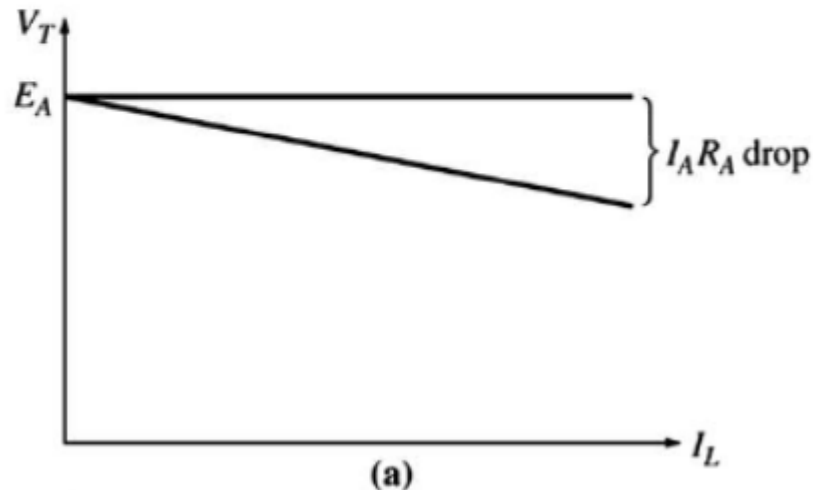
$$R_A = 0.1053 \, \Omega, R_{\text{interpoles}} = 0.0306 \, \Omega,$$

$$R_{\text{comp windings}} = 0.0141 \, \Omega, R_F = 96.3 \, \Omega$$

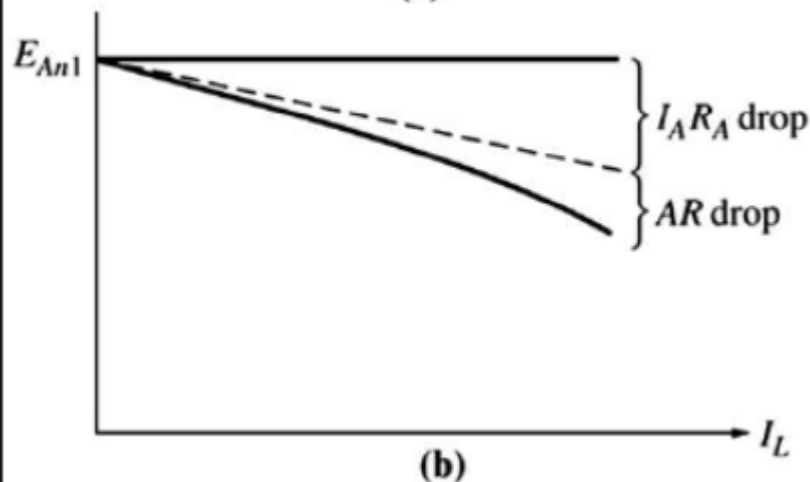
$$I_A = \frac{P}{V_T} = \frac{25000}{250} = 100 \, \text{A}$$

$$E_A = 250 + 100(0.1053 + 0.0306 + 0.0141) = 265 \, \text{V}$$

## V-I Characteristics of a Separately Excited DC Generator



(a) When load supplied  
by generator  $\uparrow$   
 $\Rightarrow I_L$  (hence  $I_A$ )  $\uparrow$   
 $\Rightarrow I_A R_A \uparrow \Rightarrow V_T \downarrow$



(b) Armature reaction present  
(no compensating windings)  
 $\Rightarrow$  flux weakening  
 $\Rightarrow E_A = K(\phi \downarrow)\omega_m \downarrow$   
 $\Rightarrow V_T \downarrow$  further

## Control of Terminal Voltage of a Separately Excited DC Generator

The terminal voltage can be controlled by changing the internal generated voltage of the machine, as

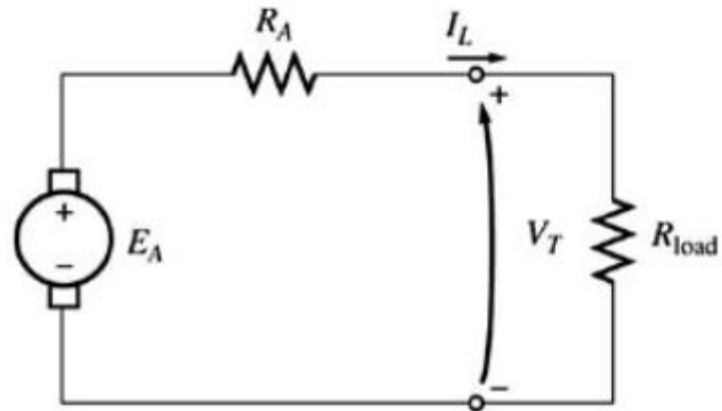
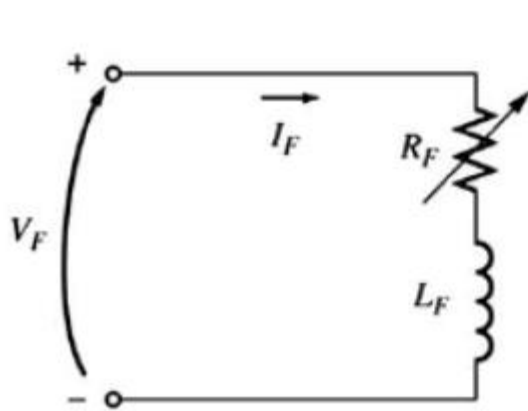
$$V_T = E_A - I_A R_A \Rightarrow \text{If } E_A \uparrow \Rightarrow V_T \uparrow$$

Since  $E_A = K\phi\omega_m$ , there are 2 ways to control  $E_A$ .

1. Change  $\omega_m$  – if  $\omega_m \uparrow \Rightarrow E_A \uparrow \Rightarrow V_T \uparrow$

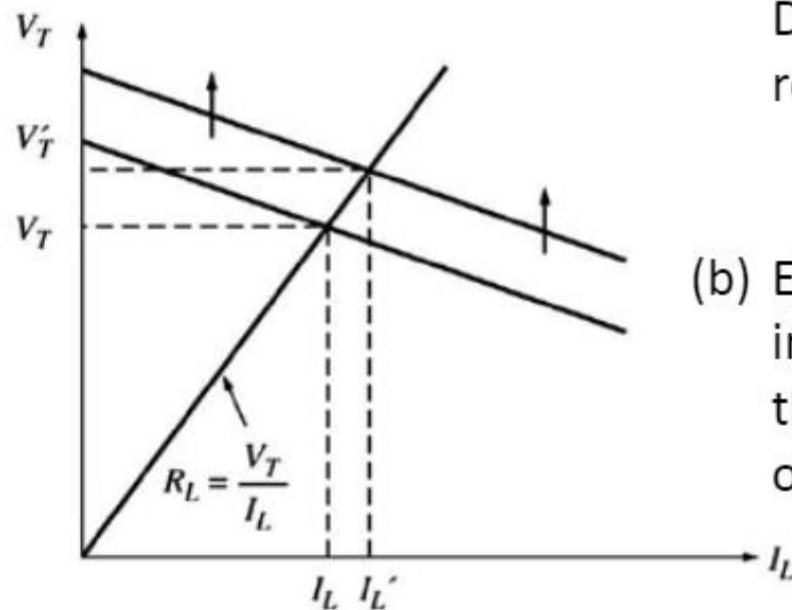
2. Change  $I_F$  – if  $R_F \downarrow \Rightarrow I_F \uparrow \Rightarrow \phi \uparrow \Rightarrow E_A \uparrow \Rightarrow V_T \uparrow$

In many applications, the speed range of the prime mover is quite limited, so the terminal voltage is most commonly controlled by changing the field current.



(a)

(a) A separately excited DC generator with a resistive load.



(b)

(b) Effect of a decrease in field resistance on the output voltage of the DC generator.

## **Nonlinear Analysis of a Separately Excited DC Generator**

- If a machine has armature reaction AR, its flux will be reduced with each increase in load, causing the internal generated voltage to decrease.
- The output voltage of the generator with AR is determined using the magnetisation curve of the generator.
- The total mmf in the generator is the field circuit mmf less the mmf due to AR.

$$mmf_{net} = N_F I_F^* = N_F I_F - mmf_{AR}.$$

Then, 
$$I_F^* = I_F - \frac{mmf_{AR}}{N_F}$$

where  $I_F^*$  = equivalent field current that will produce the same output voltage as the combination of all the mmfs in the machine.

The resulting voltage  $E_{A0}$  can then be determined by locating that equivalent field current on the magnetisation curve.

The difference between the speed  $n_0$  at which the magnetisation curve was obtained and the actual speed  $n_m$  of the generator

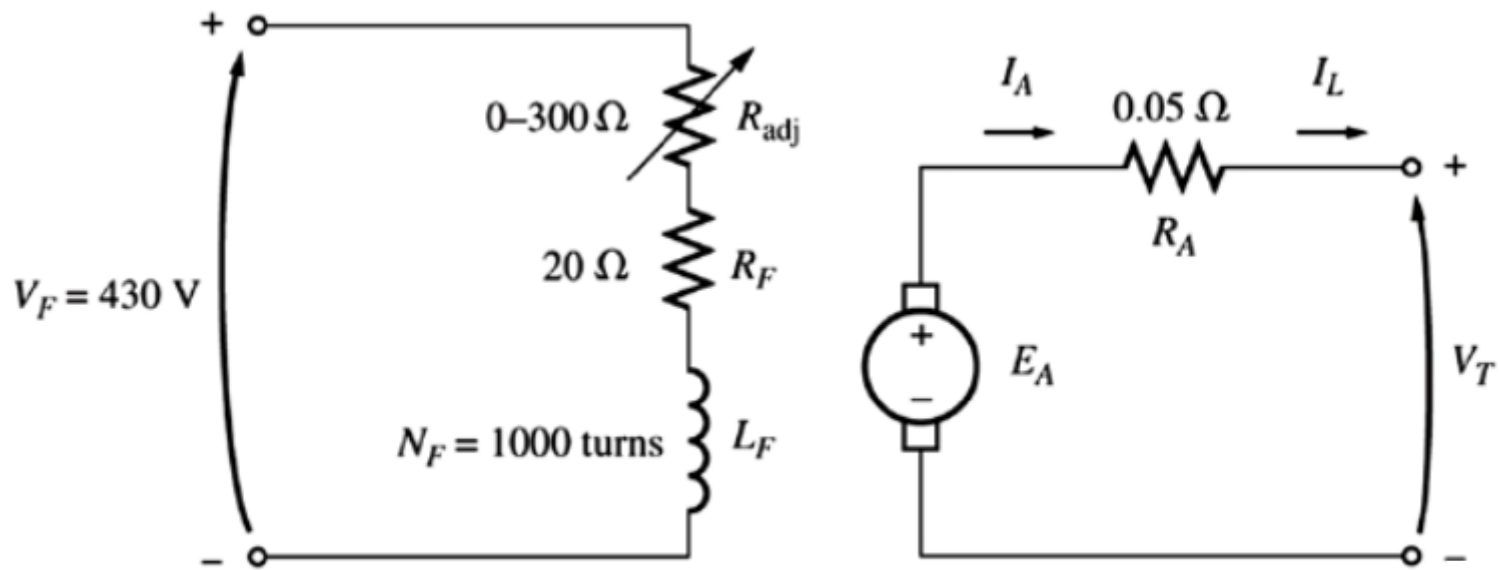
must be taken into account using 
$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0}.$$

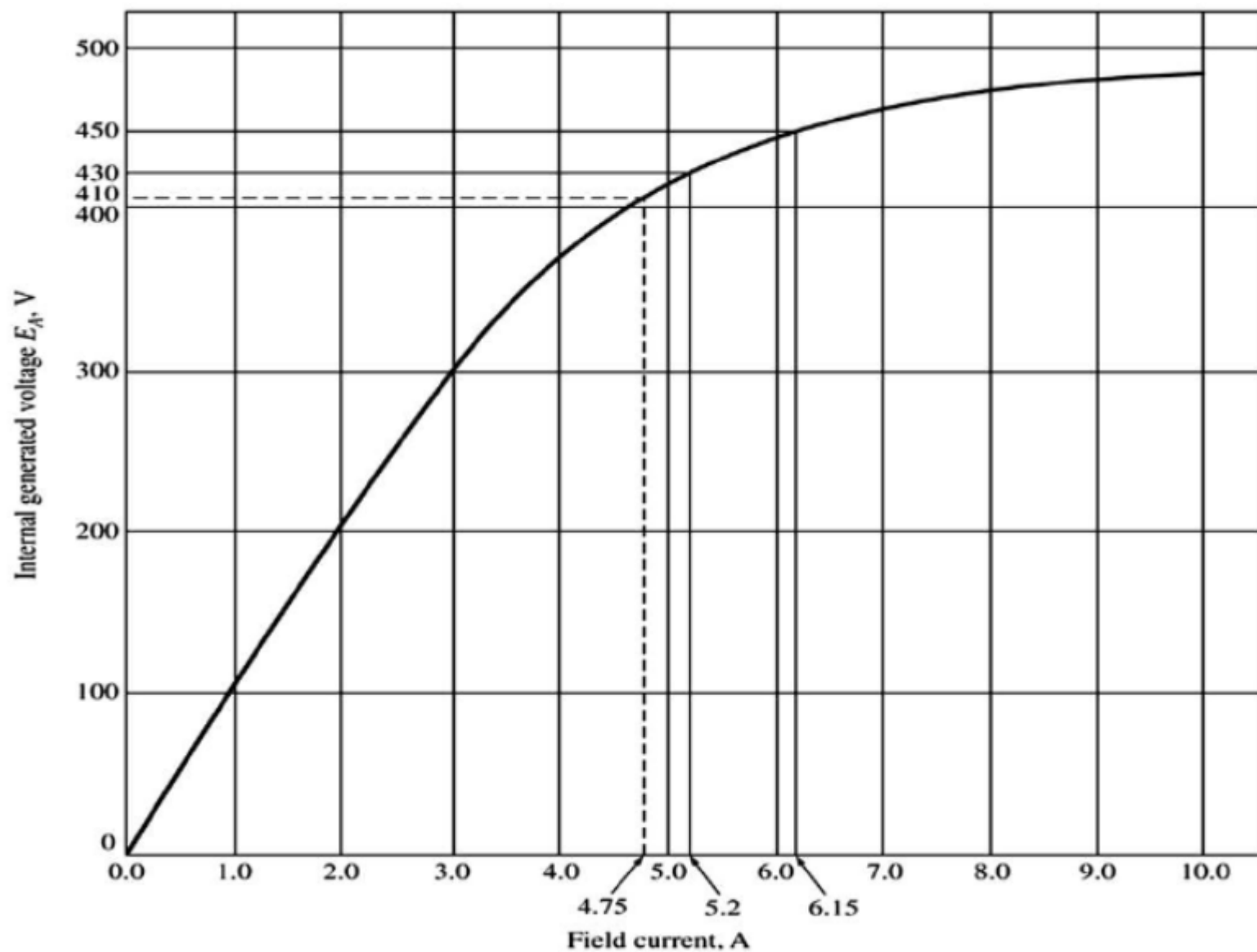
## Example on Separately Excited DC generator

A separately excited DC generator is rated at 172-kW, 430-V, 400-A and 1800-r/min. The magnetisation curve is as shown and the machine characteristics are :

$$R_A = 0.05\Omega, V_F = 430\text{V}, R_F = 20\Omega, R_{adj} = 0 \text{ to } 300\Omega,$$

$$N_F = 1000 \text{ turns per pole}$$





Note: When the field current is zero,  $E_A$  is about 3 V.

- (a) If the variable resistor in the generator's field circuit is adjusted to 63 ohms and the generator's prime mover is driving it at 1600 r/min, what is the generator's no load-terminal voltage?
- (b) What would its terminal voltage be if a 360A load were connected to its terminals? Assume that the generator has compensating windings.
- (c) What would its terminal voltage be if a 360A load were connected to its terminals but the generator does not have compensating windings? Assume that its armature reaction at this load is 450 A-turns.
- (d) For the conditions in (b), what adjustment could be made to the generator to restore its terminal voltage to the value found in part (a)?
- (e) How much field current would be needed to restore the terminal voltage to its no-load value? (Assume that the machine has compensating windings) What is the required value of the variable resistor to accomplish this?

$$(a) \quad I_F = \frac{V_T}{R_F} = \frac{430}{(20 + 63)} = 5.2 \text{ A}$$

From the magnetisation curve, this much current would produce an  $E_{A0} = 430 \text{ V}$  at a speed of  $n_0 = 1800 \text{ rpm}$ . Since the generator is rotating at  $n_m = 1600 \text{ rpm}$ , its internal generated voltage will be

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \Rightarrow E_A = 382 \text{ V}$$

Since  $V_T = E_A$  at no-load conditions, the generator's no-load terminal voltage is  $V_{T_{nl}} = 382 \text{ V}$ .

(b) There is no AR in the machine.

If a 360-A load were connected to the generator's terminals, the terminal voltage of the generator would be

$$V_T = E_A - R_A I_A = 382 - (360)(0.05) = 364 \text{ V}$$

(c) If a 360-A load were connected to the generator's terminals and the generator has 450 A-turns of AR, then

$$I_F^* = I_F - \frac{mmf_{AR}}{N_F} = 5.2 - \frac{450}{1000} = 4.75 \text{ A}$$

From the magnetisation curve, this much current would produce an  $E_{A0} = 410$  V at 1800 rpm, so the internal generated voltage at 1600 rpm would be

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \Rightarrow E_A = 364 \text{ V}$$

Therefore, the terminal voltage of the generator would be

$$V_T = E_A - R_A I_A = 364 - (360)(0.05) = 346 \text{ V}$$

It is lower than before due to the armature reaction.

(d) Increase  $E_A$  by decreasing  $R_{adj}$  so as to increase  $I_F$ .

(e) For  $V_T = 382 \text{ V}$ , the required  $E_A$  is

$$E_A = V_T + R_A I_A = 382 + (360)(0.05) = 400 \text{ V}$$

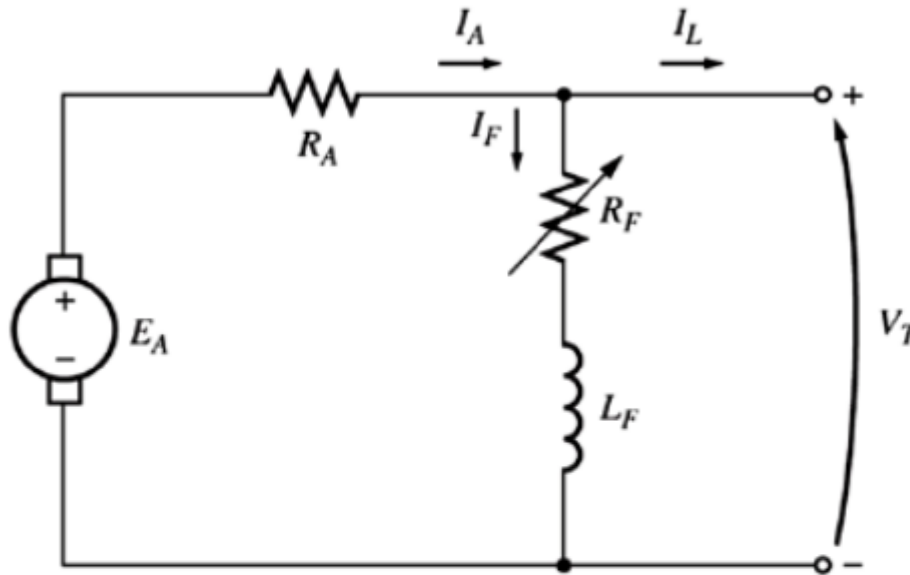
To get  $E_A = 400 \text{ V}$  at  $n_m = 1600 \text{ rpm}$ , the internal generated equivalent voltage at 1800 rpm would be

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \Rightarrow E_{A0} = 450 \text{ V}$$

From the magnetisation curve, this voltage would require a field current  $I_F = 6.15 \text{ A}$ , so that

$$20 + R_{adj} = \frac{430}{6.15} \Rightarrow R_{adj} = 50 \Omega$$

## The Shunt DC Generator



$$\begin{aligned} I_A &= I_F + I_L, \\ V_T &= E_A - I_A R_A, \\ I_F &= \frac{V_T}{R_F} \end{aligned}$$

Equivalent circuit of a shunt DC generator

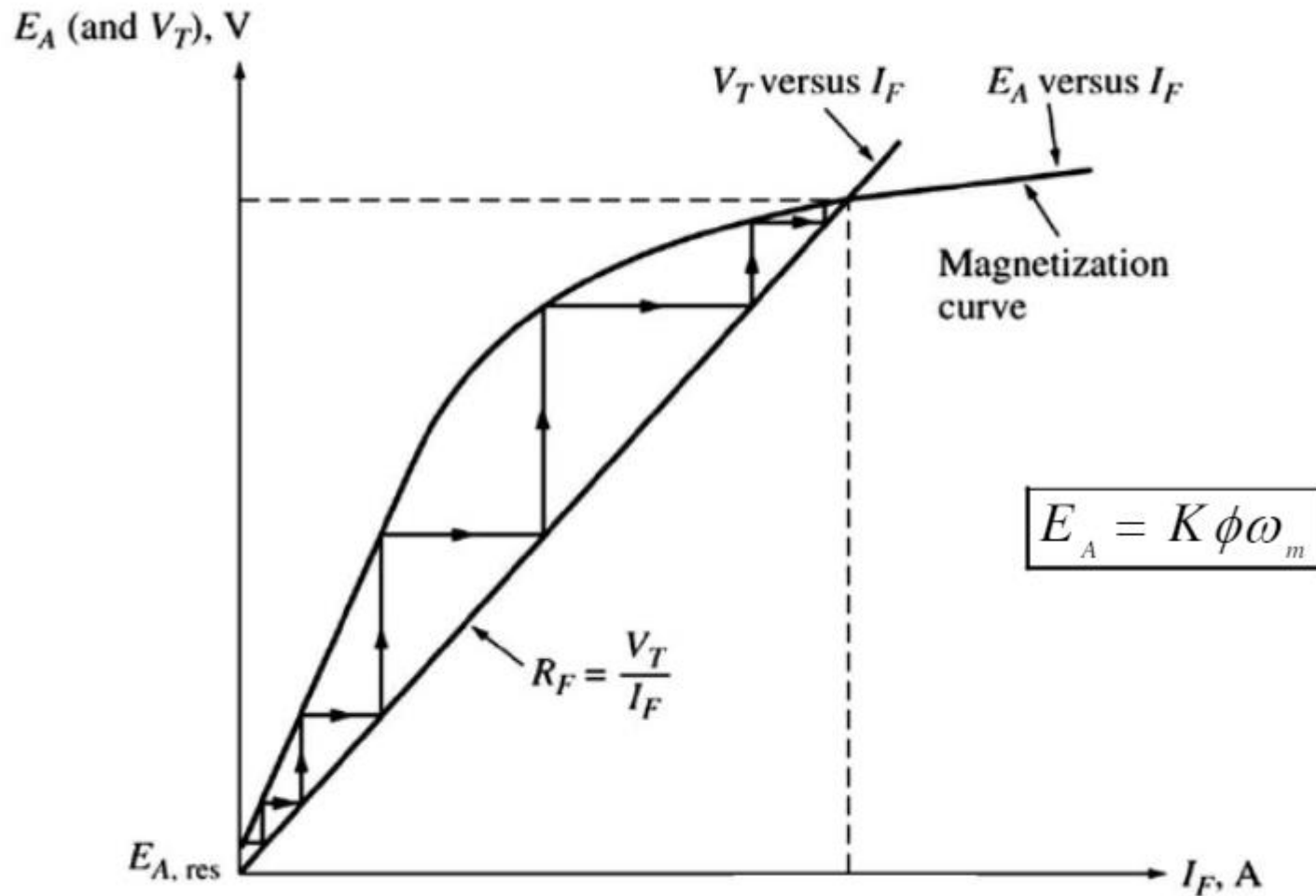
The generator supplies its own field current by having its field winding connected directly across the terminals of the machine.

The armature current of the machine supplies both the field circuit and the load attached to the machine.

This type of generator has an advantage over the separately excited generator in that no external power supply is required for the field circuit.

If the generator supplies its own field current, how does it get the initial field flux when it is first turned on?

## Voltage Build Up in a Shunt DC Generator



Voltage build up on starting in a shunt DC generator

The voltage build up depends on the presence of a residual flux  $\phi_{res}$  in the machine. When the generator first starts to turn with no load connected to it, an internal voltage given by  $E_A = K \phi_{res} \omega_m$  will be generated and a small  $V_T$  appears across the field windings and  $I_F$  flows which produces a mmf and the flux increases which in turns increases  $E_A$ , etc. It is the effect of the magnetic saturation in the machine which eventually limits the terminal voltage of the generator.

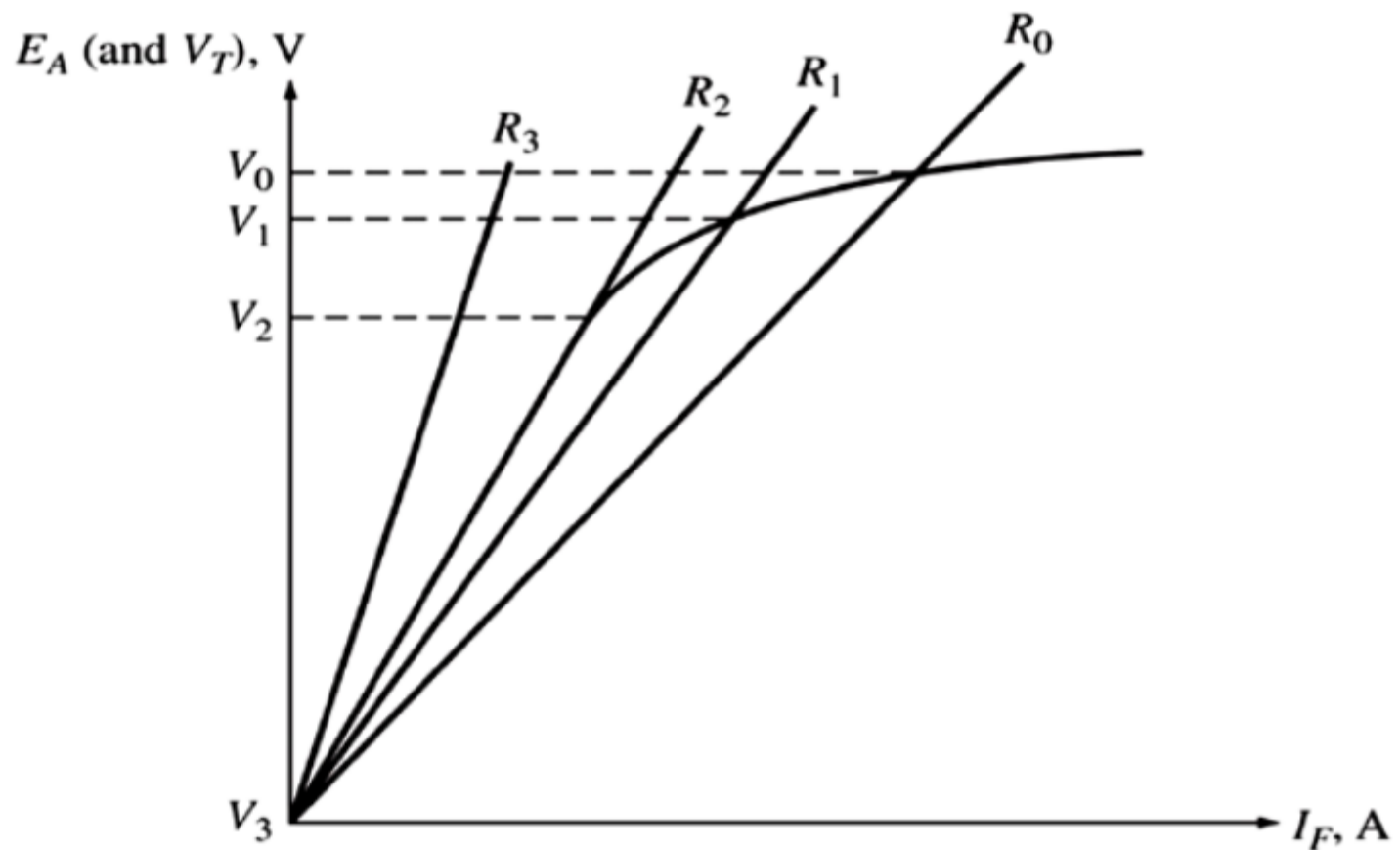
The field resistance should be less than  $R_{critical}$  for voltage build up to occurs.

Normally, the shunt DC generator's voltage will build up to the point where the magnetisation curve intersects the field resistance line, giving the no-load terminal voltage.

The field resistance should be less than  $R_{critical}$  for voltage build up to occur.

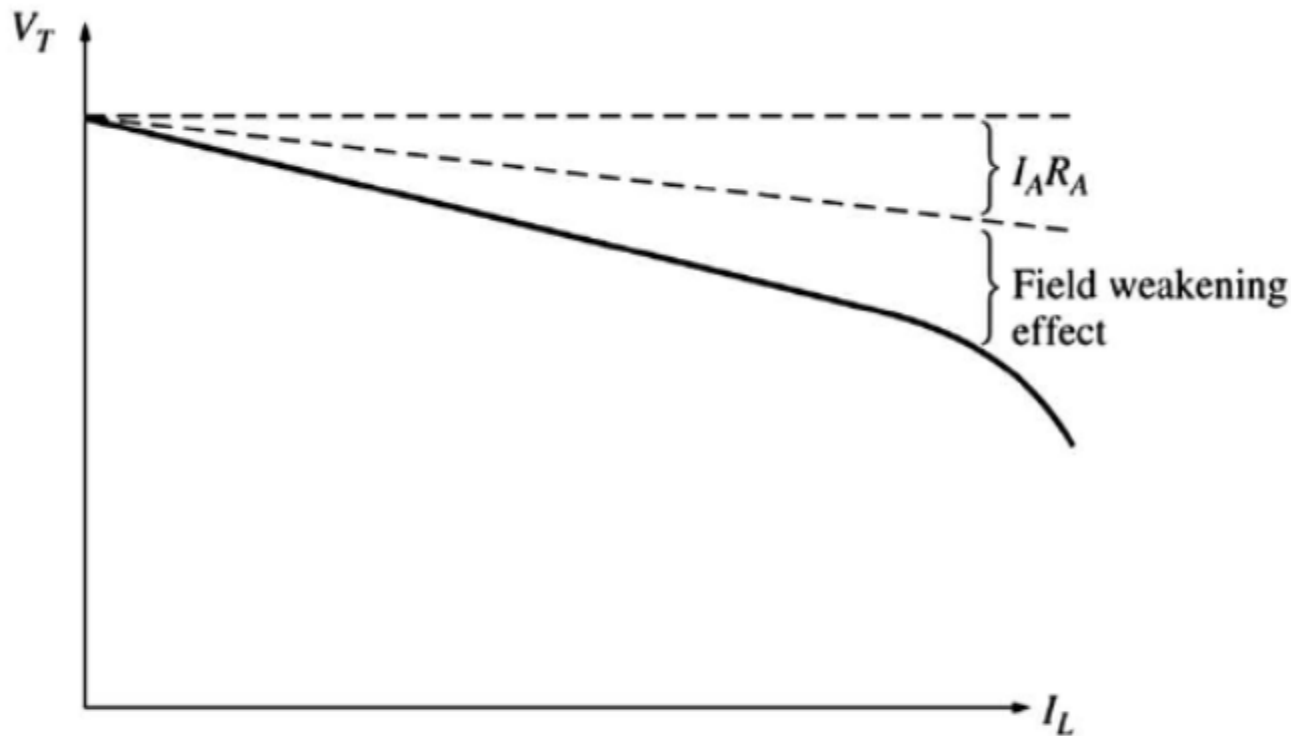
If  $R_F$  exceeds  $R_{critical}$ , then the steady-state operating voltage is essentially at the residual level, and it never builds up.

The solution to this problem is to reduce  $R_F$ .



The effect of the field resistance on the no-load terminal voltage. If  $R_F > R_2 = R_{\text{critical}}$ , then the generator's voltage will never build up.

## V-I characteristics of a Shunt DC Generator



When load on the generator  $\uparrow \Rightarrow I_L$  (hence  $I_A$ )  $\uparrow$   
 $\Rightarrow I_A R_A \uparrow \Rightarrow V_T \downarrow$ . When  $V_T \downarrow \Rightarrow I_F \downarrow \Rightarrow \phi \downarrow \Rightarrow E_A \downarrow$   
 $\Rightarrow V_T \downarrow$  further

## **Voltage Control of a Shunt DC Generator**

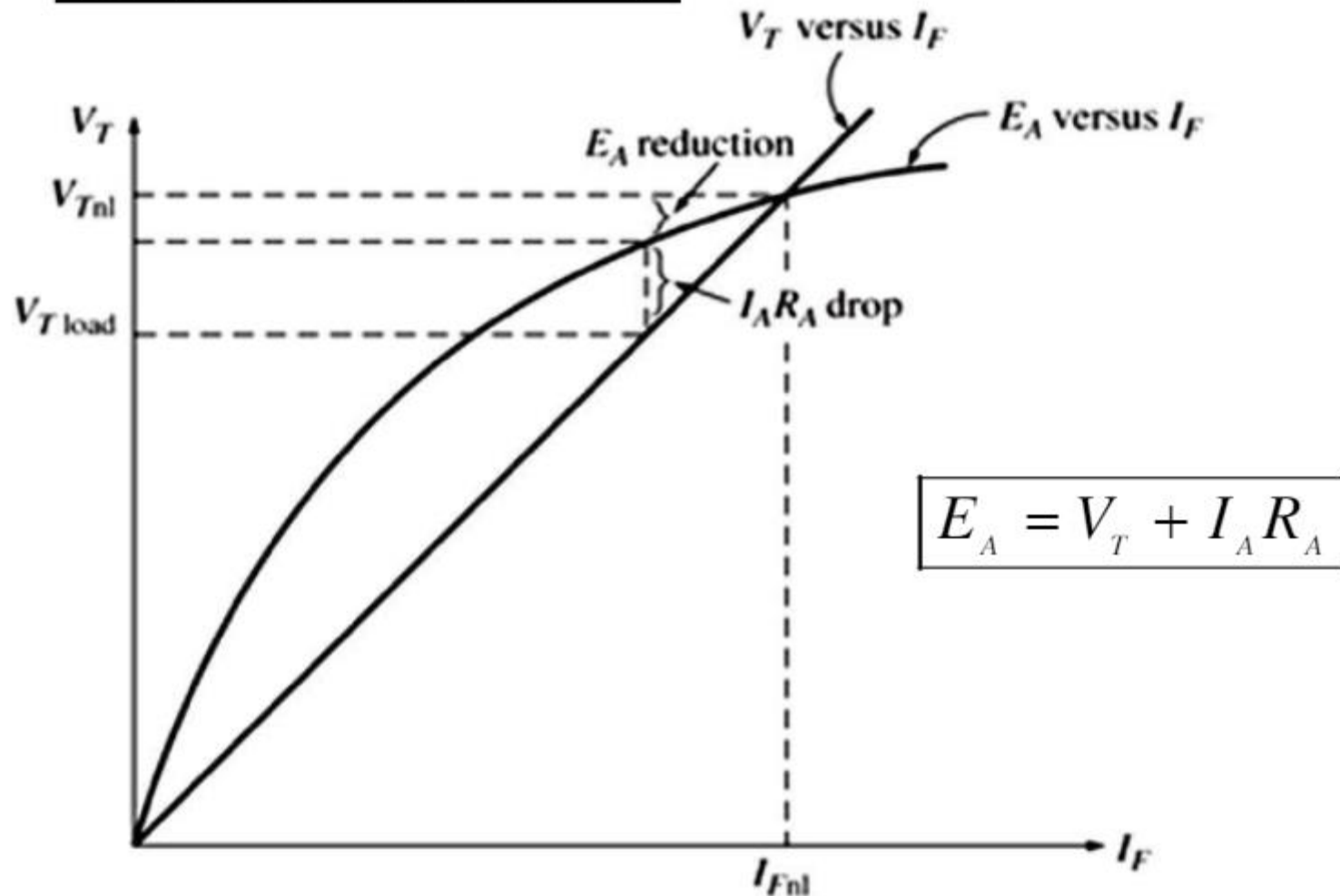
As with the separately excited DC generator, there are two ways to control the voltage of a shunt DC generator:

1. Change the shaft speed of the generator.
2. Change the total field resistance of the generator, thus changing the field current – this method is widely used.

## **The Analysis of Shunt DC Generators**

- The analysis of a shunt DC generator is more complicated than the separately excited generator, because the field current in the machine depends directly on the machine's own output voltage.
- Shunt DC generators with armature reaction will further complicate the analysis and will not be discussed in this course.

## Graphical analysis of a shunt DC generator with compensating windings



The field resistance  $R_F = \frac{V_F}{I_F}$  is shown by the straight line laid over the magnetisation curve.

At no load,  $V_T \simeq E_A$  and the generator operates at the voltage where the  $R_F$  line intersects the magnetisation curve.

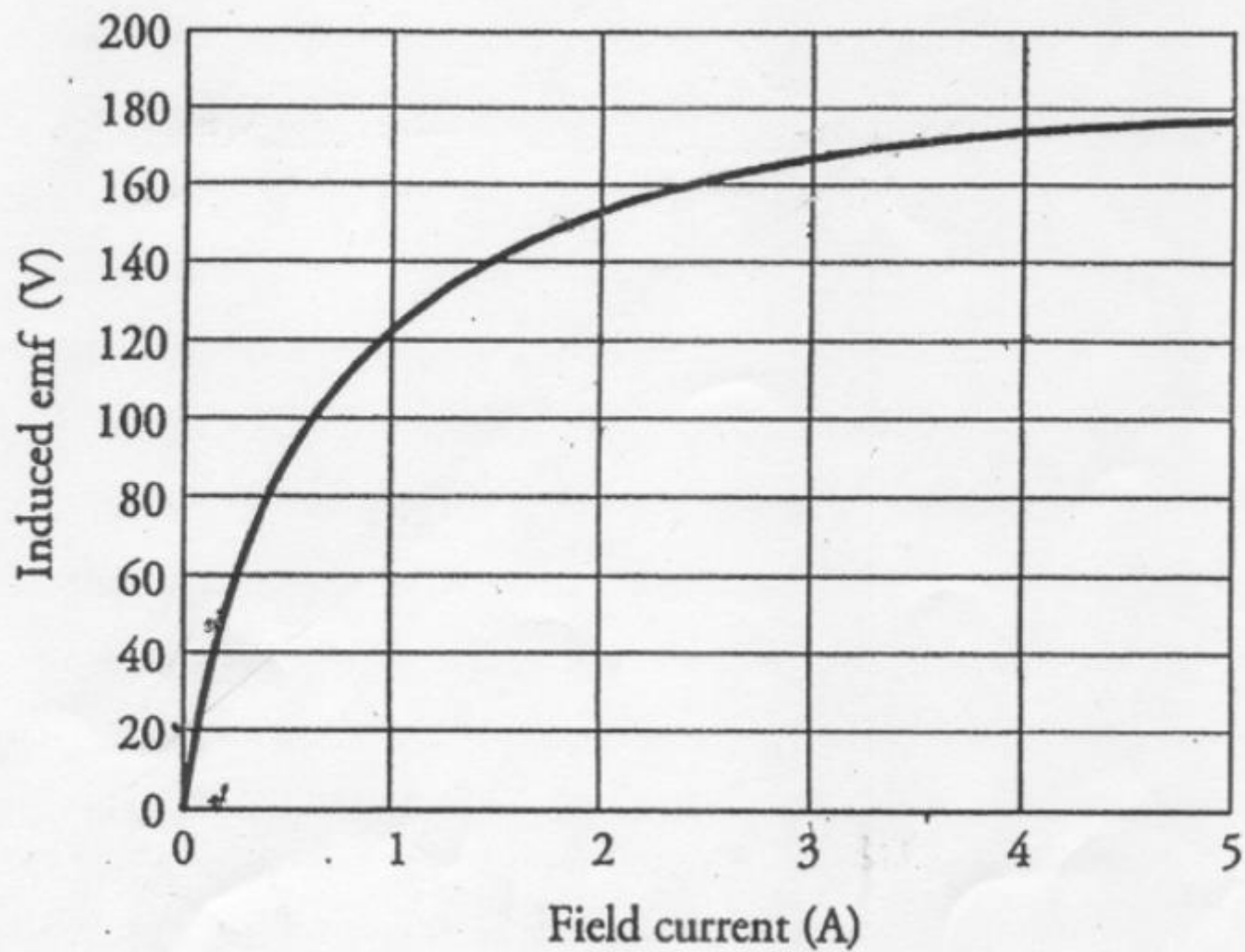
As  $E_A = V_T + I_A R_A$ , to find the terminal voltage for a given load, just determine the  $I_A R_A$  drop and locate the place on the graph where that drop fits exactly between the  $E_A$  line and the  $V_T$  line.

If there are two possible places on the curve where the  $I_A R_A$  drop will fit exactly, the one nearer the no-load voltage will represent a normal operating point.

### **Example on a DC Shunt Generator**

A DC shunt generator is driven at 1200 rpm. The magnetization curve at 1200 rpm is shown and the armature and field resistances are 0.2 ohm and 30 ohms, respectively. Neglect armature reaction.

- (i) An external rheostat is inserted in series with the field winding and adjusted to give a no-load voltage of 170 V. What is the value of the rheostat setting?
- (ii) If the generator supplies an armature current of 103.5A, determine the armature terminal voltage with the rheostat setting obtained in (i).



(a) From the magnetisation curve, the field current corresponding to a no-load terminal voltage of 170 V is 3.5 A. The total field-circuit resistance is

$$R_F = \frac{170}{3.5} = 48.57 \, \Omega$$

Thus, the external rheostat setting in the field-winding circuit is

$$R_{ext} = 48.57 - 30 = 18.57 \, \Omega$$

(b) For  $I_A = 103.5\text{A}$ ,

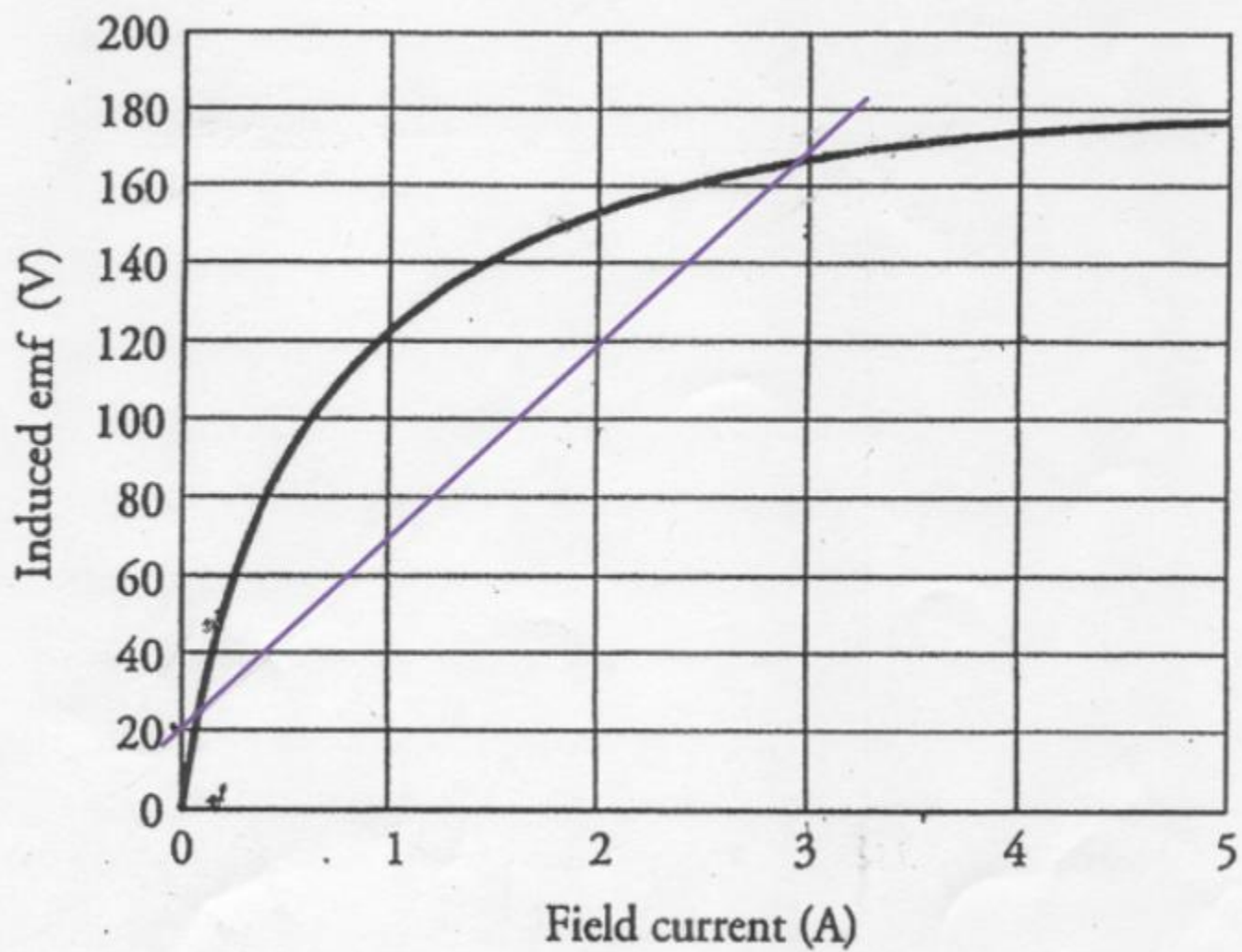
$$V_T = I_F (48.57) = E_A - (103.5)(0.2)$$

$$\Rightarrow E_A = 48.57 I_F + 20.7$$

Superimposing the above equation onto the magnetisation curve yields the intersection point

$$I_F = 3\text{A}, E_A = 166\text{V}$$

$$\Rightarrow V_T = 48.57 I_F = 145.71\text{V}$$



## Losses and Efficiency of a DC Generator

Consider a shunt DC generator as an example:

Power output,  $P_{out} = V_T I_L$

Power input,  $P_{in} (mech) = T_{in} \omega_m$  where  $T_{in}$  is the prime mover's torque.

$$P_{dev} = E_A I_A$$

$$P_{in} (mech) = P_{dev} + P_{rot}$$

$$\text{Generator Efficiency, } \eta = \frac{P_{out}}{P_{in}} \times 100\%$$

Note : For separately-excited DC generator,

$$P_{in} = T_{in} \omega_m + V_F I_F$$

## Summary

- DC generators are DC machines used as generators. The different types differ in the manner in which their field fluxes are derived.
- Since nonlinear effects are at the heart of the generator's operation, the output voltages of DC generators are determined graphically.